

EQUATION, RATION AND PROPORTION

Q1



Directions for questions 1 and 2: Select the correct alternative from the given choices.

If the sum of three consecutive numbers is 78, find the product of the least and the greatest of these numbers.

☐ a) 655

☐ b) 625

☒ c) 675 (Your answer - ✓)

☐ d) 728

Q2



Directions for questions 1 and 2: Select the correct alternative from the given choices.

Find

$$\frac{a^3 - 2a^2b + 2ab^2 - 2b^3}{a^3 - 3b^3}$$

if $a : b = 3 : 2$.

☐ a) 1

☐ b) $4/3$

☒ c) $-1/3$ (Your answer - ✓)

☐ d) -3

Q3



Directions for question 3: Type in your answer in the input box provided below the question.

What number is to be added to each of the numbers 4, 7, 14 and 22, so that they become the terms of a proportion?

Your Answer: 2 ✓

Q4



Directions for question 4: Select the correct alternative from the given choices.

The cube of a number A is proportional to the fourth power of another number B. If $B = 8$, then $A = 8$. If $B = 3$, what is the value of A?

- ☐ a) $3\sqrt{3}$
- ☐ b) $\frac{3}{2}\sqrt[3]{3}$
- ☐ c) $\left(\frac{27}{8}\right)^{\frac{1}{3}}$
- ☐ d) $\sqrt[3]{9}$

Q5



Directions for question 5: Type in your answer in the input box provided below the question.

Four persons A, B, C and D have a total amount of Rs.16,000 among them. B has an amount equal to the amount with A and D together. If C has twice as much amount as that of B, then find the amount with C (in Rs.).

Your Answer: 8000 ✓

Directions for question 6: Select the correct alternative from the given choices.

Three brothers – A, B and C divided a property among themselves and their mother in the ratio

3 : 3 : 3 : 4. If their mother got Rs.5,000 more than each son, how much did all the three brothers together get?

- ☐ a) Rs.15,000
- ☐ b) Rs.65,000
- ☐ c) Rs.30,000
- ☐ d) Rs.45,000

Solution:

The shares of A, B, C and their mother M are in the ratio 3 : 3 : 3 : 4. The mother gets 1 part more than any of the sons, and this one part is 5000. A, B, C together get 9 parts. These 9 parts are ₹45000. Choice (D)

Directions for question 7: Type in your answer in the input box provided below the question.

A person is 15 times as old as his son. Five years later, he will be five times as old as his son will be. What is the present age of the person (in years)?

Your Answer: 30 ✓

Directions for question 8: Select the correct alternative from the given choices.

The frequency of vibrations (n) in a wire is inversely proportional to its length. If the number of vibrations per second is 240 when the length is 100 cm, find the number of vibrations per second when the length is 120 cm.

- ☐ a) 50
- ☐ b) 500
- ☐ c) 120
- ☐ d) 200

Solution:

$$n \propto \frac{1}{L} \Rightarrow nL = k$$

When $L = 100$ cm, $n = 240/\text{sec}$

$$\therefore k = 24000 \text{ cm/sec}$$

$$\therefore \text{When } L = 120 \text{ cm, } n = \frac{24000}{120} \frac{1}{\text{sec}} = 200/\text{sec}$$

Choice (D)

Directions for question 9: Type in your answer in the input box provided below the question.

A man is employed for 30 days on the condition that he receives Rs.100 for each day he works and loses Rs.25 for each day he is absent. He gets Rs.2,000 in all. For how many days was he absent?

Directions for questions 10 and 11: Select the correct alternative from the given choices.

Solve the following equations for x and y .

$$4x - 3y = 7, 3x + 2y = 18.$$

☐ a) 3, 2

☐ b) 3, 4

☒ c) 4, 3 (Your answer - ✓)

☐ d) 5, 3

Directions for questions 10 and 11: Select the correct alternative from the given choices.

A tank is filled to $\frac{4}{5}$ th of its capacity. If twelve litres of water is drawn out, the tank now contains $\frac{1}{5}$ th of its total capacity. What is the capacity of the tank?

☐ a) 40 litres

☐ b) 25 litres

☒ c) 20 litres (Your answer - ✓)

☐ d) 15 litres

Solution:

The data is tabulated below

	10 paise	20 paise	50 paise
Number	a	2a	4a
Value	10a	40a	200a

As $10a + 40a + 200a = 1250$, $a = 5$.

Ans : (5)

Directions for question 13: Select the correct alternative from the given choices.

In a two – digit number, the digit in the units place is twice that in the tens place and if two is subtracted from the sum of the digits, the result is equal to $\frac{1}{6}$ th of the number. What is the number?

☐ a) 36

☐ b) 84

☐ c) 21

☒ d) 24 (Your answer - ✓)

Directions for question 14: Type in your answer in the input box provided below the question.

168 men can dig a 3150 m long canal in 36 days. How many men are required to dig a 3900 m long canal in 24 days?

Solution:

The data is tabulated below

Men	no. of days	length of canal
168	36	3150 m
N	24	3900 m

$$\frac{(168)(36)}{3150} = \frac{n(24)}{3900} \Rightarrow n = 168 \left(\frac{3}{2} \right) \left(\frac{390}{315} \right) = 312$$

Ans : (312)

Directions for question 15: Select the correct alternative from the given choices.

If nine is added to both the numerator and the denominator of a certain fraction, the fraction becomes $\frac{3}{4}$. If one is subtracted from both the numerator and the denominator, it becomes $\frac{1}{2}$. The fraction is

☐ a) $\frac{13}{20}$

☐ b) $\frac{6}{15}$

☒ c) $\frac{6}{11}$ (Your answer - ✓)

☐ d) $\frac{10}{13}$